

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 40%

Annexure A

ANALYTICAL PROBLEM SOLVING

1. A university network is assigned the IP address block 192.168.20.0/24 for its internal communication. The network administrator has decided to divide this network into subnets for three different labs using fixed subnet masks. Lab A is assigned a /26 subnet, Lab B is assigned a /27 subnet, and Lab C is assigned a /28 subnet. Based on the given subnet masks, apply subnetting concepts to determine the network address of each lab. Further, identify the valid range of host IP addresses and the broadcast address for each subnet. Finally, calculate the total number of usable host IP addresses available in each lab's subnet.
2. A network administrator has configured two systems with the following details: System 1 has IP address 192.168.1.130 with subnet mask 255.255.255.192, and System 2 has IP address 192.168.1.190 with subnet mask 255.255.255.192. Using the given configuration, apply subnetting techniques to determine the subnet range, network address, and broadcast address for each system. Identify the valid host IP range for both subnets based on the given subnet

mask. Calculate whether each IP address falls within its respective subnet range. Also, determine the number of usable host addresses available in each subnet.

3. An organization has been allocated the IP block **172.16.0.0/16** for its internal network. The network administrator has already decided the subnet masks for each department as follows: HR → /25, Sales → /26, IT → /27, and Admin → /28. Using the given subnet masks, apply subnetting techniques to assign suitable subnet addresses for each department. Determine the network address, valid host IP range, and broadcast address for each subnet. Based on the allocation, calculate the number of usable host addresses in each department. Finally, identify the remaining unused IP address range within the given network block.

4. A company uses the IP address **172.16.10.0/24** and has implemented subnetting using a **/27 subnet mask** for its internal departments. Using the given subnet mask, apply subnetting techniques to calculate the total number of subnets created and the number of usable hosts per subnet. Determine the subnet to which the IP address **172.16.10.78** belongs. Identify the network address and broadcast address for that subnet. Also, determine whether the IP address **172.16.10.95** falls within the same subnet range based on the given configuration.

5. A network uses the Internet Protocol version 6 address **2001:0db8:0000:0000:0000:ff00:0042:8329**. Apply IPv6 addressing rules to compress the given address into its shortest form and then expand the compressed address back to its full notation. Using a prefix length of **/64**, determine the

network prefix of the given address. Also, calculate the total number of possible host addresses available within this network based on the given prefix length.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

Analytical Problem Solving – Computer Networks

1. Subnetting 192.168.20.0/24

Given network: **192.168.20.0/24**

The administrator allocates:

- Lab A → **/26**
- Lab B → **/27**
- Lab C → **/28**

Use **VLSM allocation in order**.

Lab A (/26)

- Subnet mask: **255.255.255.192**
- Block size = **64**

Item	Value
Network Address	192.168.20.0
First Host	192.168.20.1
Last Host	192.168.20.62
Broadcast Address	192.168.20.63

Usable hosts:

- Total = $2^{(32-26)} = 64$
- Usable = **64 – 2 = 62**

Lab B (/27)

Next available address after Lab A is **192.168.20.64**

- Subnet mask: **255.255.255.224**
- Block size = **32**

Item	Value
Network Address	192.168.20.64
First Host	192.168.20.65
Last Host	192.168.20.94
Broadcast Address	192.168.20.95

Usable hosts:

- Total = 32
- Usable = **30**

Lab C (/28)

Next available address after Lab B is **192.168.20.96**

- Subnet mask: **255.255.255.240**
- Block size = **16**

Item	Value
Network Address	192.168.20.96
First Host	192.168.20.97
Last Host	192.168.20.110
Broadcast Address	192.168.20.111

Usable hosts:

- Total = 16
- Usable = **14**

Final Answer

Lab	Subnet	Network	Host Range	Broadcast	Usable Hosts
A	/26	192.168.20.0	1 – 62	192.168.20.63	62
B	/27	192.168.20.64	65 – 94	192.168.20.95	30
C	/28	192.168.20.96	97 – 110	192.168.20.111	14

2. Analyze Two Systems

Subnet mask: **255.255.255.192**

Block size = **64**

Possible subnets:

- 192.168.1.0 – 63
- 192.168.1.64 – 127
- **192.168.1.128 – 191**
- 192.168.1.192 – 255

System 1

IP: **192.168.1.130**

Belongs to subnet **128 – 191**

Item	Value
Network	192.168.1.128
First Host	192.168.1.129
Last Host	192.168.1.190
Broadcast	192.168.1.191

Check:

- 130 lies between **129 and 190**

- **Valid host address**

System 2

IP: **192.168.1.190**

Same subnet **128 – 191**

Item	Value
Network	192.168.1.128
First Host	192.168.1.129
Last Host	192.168.1.190
Broadcast	192.168.1.191

Check:

- 190 is the **last usable host**
- **Valid host address**

Usable Hosts

For **/26**:

- Total addresses = 64
- Usable hosts = **62**

Conclusion

System	IP	Network	Broadcast	Valid?
1	192.168.1.130	192.168.1.128	192.168.1.191	Yes
2	192.168.1.190	192.168.1.128	192.168.1.191	Yes

Both systems are in the **same subnet** and can communicate directly.

3. Subnetting 172.16.0.0/16

Given block: **172.16.0.0/16**

Departments:

- HR → /25
- Sales → /26
- IT → /27
- Admin → /28

Allocate sequentially.

HR (/25)

Item	Value
Network	172.16.0.0

Host Range	172.16.0.1 – 172.16.0.126
Broadcast	172.16.0.127
Usable Hosts	126

Sales (/26)

Next available: **172.16.0.128**

Item	Value
Network	172.16.0.128
Host Range	172.16.0.129 – 172.16.0.190
Broadcast	172.16.0.191
Usable Hosts	62

IT (/27)

Next available: **172.16.0.192**

Item	Value
Network	172.16.0.192
Host Range	172.16.0.193 – 172.16.0.222
Broadcast	172.16.0.223
Usable Hosts	30

Admin (/28)

Next available: **172.16.0.224**

Item	Value
Network	172.16.0.224
Host Range	172.16.0.225 – 172.16.0.238
Broadcast	172.16.0.239
Usable Hosts	14

Remaining Unused Range

Used addresses:

- **172.16.0.0 – 172.16.0.239**

Unused portion:

- **172.16.0.240 – 172.16.255.255**

Summary

Department	Network	Host Range	Broadcast	Usable
HR	172.16.0.0/25	1 – 126	172.16.0.127	126
Sales	172.16.0.128/26	129 – 190	172.16.0.191	62
IT	172.16.0.192/27	193 – 222	172.16.0.223	30
Admin	172.16.0.224/28	225 – 238	172.16.0.239	14

4. Subnetting 172.16.10.0/24 Using /27

Step 1: Number of Subnets

Original network = /24

New subnet mask = /27

Borrowed bits = $27 - 24 = 3$

Number of subnets:

- $2^3 = 8$ subnets

Step 2: Usable Hosts per Subnet

Host bits = $32 - 27 = 5$

- Total addresses = $2^5 = 32$
- Usable hosts = 30

Step 3: List /27 Subnets

Subnet	Range
172.16.10.0	0 – 31
172.16.10.32	32 – 63
172.16.10.64	64 – 95
172.16.10.96	96 – 127
172.16.10.128	128 – 159
172.16.10.160	160 – 191
172.16.10.192	192 – 223
172.16.10.224	224 – 255

Step 4: IP 172.16.10.78

78 falls in **64 – 95**

Item	Value
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Network	172.16.10.64
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Item	Value
First Host	172.16.10.65
Last Host	172.16.10.94
Broadcast	172.16.10.95

Step 5: Check 172.16.10.95

- 95 is the **broadcast address** of subnet 172.16.10.64/27.
- It is **not a usable host**.

Final Answer

Item	Result
Total Subnets	8
Usable Hosts per Subnet	30
Subnet of 172.16.10.78	172.16.10.64/27
Network Address	172.16.10.64
Broadcast Address	172.16.10.95
Is 172.16.10.95 a host in same subnet?	No – it is the broadcast address

5. IPv6 Address Analysis

Given IPv6 address:

2001:0db8:0000:0000:0000:ff00:0042:8329

Step 1: Compress the Address

Rules:

- Remove leading zeros.
- Replace consecutive 0000 groups with :: once.

Compressed form:

2001:db8::ff00:42:8329

Step 2: Expand Back to Full Form

Expanded form:

2001:0db8:0000:0000:0000:ff00:0042:8329

Step 3: Determine the /64 Network Prefix

Prefix length = /64

The first **64 bits** (first four groups) represent the network.

Network prefix:

2001:0db8:0000:0000::/64

Compressed:

2001:db8::/64

Step 4: Calculate Host Addresses

For /64:

- Host bits = $128 - 64 = 64$

Number of possible host addresses:

$2^{64} = 18,446,744,073,709,551,616$

Approximately **18.4 quintillion hosts**.

Final Answer

Item	Result
Original Address	2001:0db8:0000:0000:0000:ff00:0042:8329
Compressed Form	2001:db8::ff00:42:8329
Expanded Form	2001:0db8:0000:0000:0000:ff00:0042:8329
Network Prefix (/64)	2001:db8::/64
Host Portion	0000:ff00:0042:8329
Total Possible Hosts	$2^{64} = 18,446,744,073,709,551,616$

Overall Interpretation

These problems demonstrate practical application of:

- **IPv4 subnetting using VLSM**
- **Network, broadcast, and host range calculation**
- **Subnet membership verification**
- **Subnet capacity planning**
- **IPv6 compression and prefix identification**

Such calculations are essential for **efficient IP address management, network scalability, security segmentation, and reliable communication in enterprise networks**.